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ORIGINAL RESEARCH

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Abandonment of assistive products: assessing abandonment levels and factors that impact on it

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ABSTRACT

Purpose: To investigate the levels and factors that influence the abandonment of assistive products by users of a local reference rehabilitation center.

Methods: This observational study involved users who received services and assistive products provided by our center of rehabilitation. Users were identified using the records of the center and their responses about the abandonment were collected through face-to-face interviews.

Results: The abandonment level of assistive products was 19.38%. 83.5% of the users use at least one of the assistive products they have received. Rigid and folding frame wheelchairs, with and without postural support devices, as well as shower wheelchairs, presented the lowest abandonment levels, followed by canes and lower limb orthoses. Upper limb orthoses, Knee Ankle Foot Orthosis(KAFO), walkers, crutches and lower and upper limb prostheses all presented higher abandonment levels.

Conclusion: The simultaneous use of mutiple assistive products, users perception on the importance of using them, and completing the rehabilitation treatment were found to impact on the short and long-term use of products. The study offers inputs to decision making and planning for assistive technology provision in developing countries with regard to expected demand and service delivery.

➤ IMPLICATIONS FOR REHABILITATION

- Data about the abandonment of assistive products in Sao Paulo, Brazil, could assist informing decision making on provision and servicing of these products in similar settings.
- The strong correlation found between abandonment levels and the simultaneous use of multiple devices should be taken into account by health professionals when prescribing assistive products and providing guidance to users.
- The need for follow up on the use of assistive products after discharge from rehabilitation treatment becomes strikingly clear, as data show that completing treatment is significantly relevant when evaluating abandonment levels.
- As assistive products users' perception about the importance of using these devices is shown to be significant in explaining abandonment, it is mandatory that health and rehabilitation professionals take it into account when providing guidance and training users.

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Introduction

Only 5–15% of persons who need assistive products in low and middle income countries have access to one. However, the need is expected to grow as the prevalence of disability increases and population ageing (which is related to increased chances of using assistive devices) will probably make the use of assistive products even more common and widespread, requiring rehabiltation services and health systems attention [1,2].

The WHO [3] defines assistive technology as "a subset of health technology" and "the application of organized knowledge and skills related to assistive products, including systems and services". Assistive products are "any external product (including devices, equipment, instruments or software), especially produced or generally available, whose primary purpose of is to maintain or improve an individual's functioning and independence and thereby promote their well-being" and can also be used "to prevent impairments and secondary health conditions."

According to the World Report on Disability [4], when assistive products are appropriate to and match their users needs and environment, they enhance user's independence and social participation. However, the opposite is also true. Difficulties in the interaction between users, their assistive products and user's environment could lead to the abandonment of assistive products.

Abandoning assistive products can be related to characteristics of the system and interventions carried out by health, rehabilitation and social support services in each country; to the physical and psychological characteristics of users and to particularities of their environment; or even to technical features and quality of

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products. These factors could both be negative, such as worsening health conditions of the user, environmental barriers to the use of a product or the low quality of a given productor positive, such as improved health conditions, functional gains or enabling environments [5,6].

Assistive products abandoned for a negative reason might not only hinder user's rehabilitation and social inclusion, but also result in the waste of public resources. From the perspective of the healthcare system, not being able to recover assistive products that are provided and are no longer used results in a missed opportunity to save funds and resources. Thus, the economic leakage caused by the abandonment of assistive products results in "waste of resources, an unchanged situation in the person's disability and frustration for the users" [7].

Although persons with disabilities might need assistive technologies and products that improve social and community participation, there are still barriers to access and use and unmet needs [1,8]. The lack of access to assistive technology impacts functionality and influences social participation, a theme that has been discussed around the world [1].

The Physical and Rehabilitation Medicine Institute of the University of Sao Paulo Medical School General Hospital (IMREA-HCFMUSP) is a reference for secondary and tertiary levels of rehabilitation care for people with physical disabilities through the Brazilian Public Healthcare System in the State of Sao Paulo. In 2014, it was the first Brazilian institution to be accredited by the Commission on Accreditation of Rehabilitation Facilities (CARF) and is commited to improve the quality of services provided [9,10]. Hence, assessing the outcomes of the assistive products is important to improve the use of public resources.

Due to the importance of the abandonment levels of assistive products, as well as the lack of literature about the Brazilian context, the objective of this article is to show the levels and factors that influence the abandonment of assistive products by users of a local reference rehabilitation center.

Materials and methods

This study has been approved by the Ethics Committee of the University of Sao Paulo Medical School General Hospital, as per registers 71602317.9.0000.0068.

It draws from the database of a survey contracted by the Sao Paulo State Secretariat for the Rights of the Person with Disability to assess the satisfaction of users with services received at IMREA-HCFMUSP. Telephone calls were followed by face-to-face interviews to collect information on users satisfaction with the services provided by the Institute, which included the provision of assistive products. Interviews represented no risk to its respondents.

Nine thousand hundredand four users were identified as having received some kind of rehabilitation services in at least one of the five facilities that comprise the Institute in the city of Sao Paulo, Brazil. Face-to-face interviews were scheduled with 1390 (15%) users, who agreed to participate.

As 35.2.% of users gave up participating in the survey, 900 face-to-face interviews were carried out. Interviews were performed aiming at the participation of users themselves. However, due to communication impairments or legal restrictions (for those aged below 14 years), proxy interviews were attempted with caregivers and parents.

As the original questionnaire in Portuguese consisted of a number of questions on different aspects of rehabilitation service provision and social inclusion [11], a translated version of the questions referring to assistive products provision was also available.

Among other themes, respondents were questioned about the assistive products received, the use of assistive products, factors that influenced abandonment, difficulties in using assistive products and the importance users gave to using assistive products during their rehabilitation treatment.

About the design of the interview's questionnaire, it is important to highlight that it allows the identification, for each user, of what assistive products were provided and were still used. There were no specific considerations on the full- or partial-time use of these devices. Moreover, the survey did not apply a threshold between use and nonuse of assistive products, as it relies completely on user's perspectives.

When assessing the several reasons that can lead to abandonment, the questionnaire allowed different responses and open answers. For this study, however, answers were grouped together around four main categories, which are [12]:

i. Personal factors:
   a. User's health condition improved or worsened.
   b. The user did not get adapted to the device or did not accept it.
   c. The user was ashamed of using the device.
   d. A new disease or injury prevented the use of the device.
   e. The user lost or replaced the device.
ii. Intervention-related factors:
   a. The device was of inadequate size.
   b. The device did not meet user's needs.
   c. The device was inadequate to the intended functions.
   d. The use was suspended by medical orders.
   e. Using the device lead to discomfort or pain.
   f. The user could not wear the device.
   g. The user did not know how to use the device.
iii. Product-related factors:
   a. The device was worn out or broken.
   b. The device was under maintenance.
iv. Environmental factors:
   a. The user felt discriminated when using the device.
   b. The user faced barriers to using the device in or out of his/her home.

Data analysis

Data are presented using descriptive statistics and comparisons on their respective independent proportions and odds ratios, which use confidence intervals(CI's) of 95%. In all cases, only results that showed a $p < .05$ were deemed statistically different and significant.

This study allowed drawing the analysis on users that use only a part of the assistive products received, when it was necessary to adopt a clear threshold for use and nonuse (e.g., when calculating odds ratios), the threshold was established on the use of half of the products received, i.e, if a user uses up to half of the products he/she has received, this is scored as nonuse.

Results

The analysis demonstrates that, from the 900 initial respondents, 643 (71.44%) of them have received some kind of assistive product. Thus, every research finding and results presented next relate to the characteristics and answers of this smaller group of respondents.

The average age of users is 45.9 ± 23 years old, of which 376 (58.48%) are men and 267 (41.52%) are women. Among survey

Table 1. Number of assistive products provided, used or nonused, at the time of the interview.

	Provided		Used		Nonused	
Assistive products	*n*	%	*n*	%	*n*	%
Cane	66	4.24	55	83.33	11	16.67
Crutch	35	2.25	24	68.57	11	31.43
Folding frame wheelchair with PSD	206	13.22	174	84.47	32	15.53
Folding frame wheelchair without PSD	97	6.23	81	83.51	16	16.49
Lower limb orthosis (KAFO)	91	5.84	67	73.63	24	26.37
Lower limb orthosis (others)	299	19.19	244	81.61	55	18.39
Lower limb prosthesis	62	3.98	41	66.13	21	33.87
Mobile/Tablet	1	0.06	0	0.00	1	100.00
Rigid frame wheelchair with PSD	81	5.20	75	92.59	6	7.41
Rigid frame wheelchair without PSD	30	1.93	27	90.00	3	10.00
Shower wheelchair	235	15.08	204	86.81	31	13.19
Upper limb orthosis	288	18.49	221	76.74	67	23.26
Upper limb prosthesis	15	0.96	7	46.67	8	53.33
Walker	52	3.34	36	69.23	16	30.77
Total	1558	100	1256	80.62	302	19.38

Table 2. Number of users who have received at least one assistive product, and the status of use or abandonment.

	Number of users		Use all		Use a few		Use none	
Quantity of products received	*n*	%	*n*	%	*n*	%	*n*	%
Received 1	179	27.8	123	68.7	0	0.0	56	31.3
Received 2	173	26.9	127	73.4	21	12.1	25	14.5
Received 3	173	26.9	114	65.9	42	24.3	17	9.8
Received 4	81	12.6	63	77.8	12	14.8	6	7.4
Received 5	33	5.1	20	60.6	11	33.3	2	6.1
Received 6	3	0.5	2	66.7	1	33.3	0	0.0
Received 7	1	0.2	1	100.0	0	0.0	0	0.0
Total	643	100.00	450	70.0	87	13.5	106	16.5

respondents, 60.35% of them were users themselves, 32.5% were caregivers and 7.15% were parents or relatives.

Table 1 shows the number of assistive products provided by Instituto de Medicina Física e Reabilitação (IMREA-HCFMUSP) to these users and how many of them were still in use or were abandoned by the time of the interview. It is worth mentioning that each user might have received more than one assistive product according to his/her needs.

Considering the total number of users and the total number of assistive products provided, both used and nonused, it results that, on average, each user received 2.42 products, but only 1.95 were actually used at the time of the interview.

In general, on average, 19.38% of assistive products provided were abandoned. Nevertheless, it is important to highlight that the use of multiple devices can influence the use or abandonment of assistive products. Therefore, Table 2 presents the number of users per quantity of products received and how many of them used all, a few or none of the products received.

It is important to notice that Table 1 focus on products and how many of them were used or abandoned, while Table 2, on the other hand, focus on users and how many of them used all, a few or none of the products received.

Table 2 demonstrates that the majority of users (81.65%) received up to three devices.

The odds ratio on the use of assistive products comparing with those who have received only one product to those who have received more than one demonstrates that the first group has a smaller chance of use (0.36, CI: 0.24;0.54).

On the other hand, comparing the use of assistive products by users who have received three devices with those who have received less or more devices (i.e, all others), demonstrates that the first group has more chances to use the assistive products provided (1.72, CI: 1.05;2.81)

Odds ratios calculated comparing users who have received all other quantitites of products do not show statistically significant data. However, it is visible that chances to use increase for groups of users who received up to three products and then start diminishing for those who received four devices or more.

Table 3 shows the number of users who used or abandoned the assistive products they have received according to whether they were still on treatment or have completed it by the time of the interview.

The odds ratio calculated comparing users who were on treatment and those who have completed it demonstrated that the first group had more chances (2.25, CI: 1.48;3.42) to use the assistive products they have received.

The odds ratio comparing users that have completed rehabilitation treatment less than one year before the interview and and those who have completed treatment for longer indicates that the first group had more chances (2.74, CI: 1.62;4.63) to use the assistive products they have received.

When comparing the group who have completed the rehabilitation treatment between one and three years before the interview and the rest of the users who have completed treatment, the odds ratio shows a smaller chance (0.80, IC: 0.48;1.33) of the first group to use the assistive products they have received.

Users who have completed rehabilitation treatment for more than three years before the interview, in comparison to other users who have completed treatment, have even smaller chances (0.24, IC: 0.12;0.46) to use the assistive products they have received.

Apart from short- and long-term use, which can be analyzed based on the use or abandonment of assistive products during or after completing rehabilitation treatment, the perception of the user about the use of assistive products during treatment is equally important when assessing their adherence to the use of assistive products. Table 4 shows the data of users according to their perception about the use and the actual use of assistive products at the time of the interview.

When comparing users that believe the use of assistive products during rehabilitation treatment is important or really important to those who believe the use is not important at all, is only a bit important or who are indifferent to it, the odds ratio shows that the first group had higher chances (3.84, IC: 1.69;8.73) to use

Table 3. Status of users according to whether they were still on treatment or have completed it at the time of the interview.

User status	Number of users	Use	Nonused
On treatment	309	270	39
Treatment completed	334	252	82
Treatment completed less than one year before the interview	167	141	26
Treatment completed between one and three years before the interview	125	91	34
Treatment completed for more than three years before the interview	42	20	22

Table 4. Perception of users about the importance of using assistive products during treatment.

Perception of users about the importance of use	*n*	%	Use	Nonused
Not important at all	3	0.47	12	12
A bit important	4	0.62		
Indifferent	17	2.64		
Important	147	22.86	491	128
Really important	472	73.41		

the asssitive products they received, despite the significant difference between the sizes of each group.

Nevertheless, reasons to stop using an assistive product are not only personal such as individual perceptions on the importance of its usebut could also be related to several other factors, as previously mentioned. Table 5 presents the main factors appointed by users for the abandonment of each category of assistive products, highlighting that users could indicate more than one reason to abandon the assistive products they have received.

In Table 5, reasons to abandon an assistive products are also divided between positive and negative factors, indicating that assistive products can be abandoned due to improving health and functioning conditions. When listing the factors that contributed to abandoning an assistive product, categories are listed in decreasing order.

Discussion

Access to assistive technology and products is a human rights guarantee to which every signatory country of the United Nations Convention on the Rights of Persons with Disabilities, among which is Brazil, has committed. The Convention reiterates the importance of international cooperation to improve access to and the development of assistive technology. Thus, this study identifies the abandonment levels of assistive products and the factors that impact them among users of a local reference rehabilitation center in Sao Paulo, Brazil.

With reference to the users of assistive products, this study, as well as Federici et al. [5] demonstrates, a predominantly male population (58.48%). However, information on the prevalence of physical disabilities in Brazil, according to the 2010 National Census, shows that they are more prevalent among women (62.46% in Brazil and 63.46% in Sao Paulo) than among men [13]. This might imply that men have a privileged access to mobility-related assistive products, which may be related to their greater inclusion in the labor market and social inclusion in comparison to women in the same situation.

The study by Federici et al. [5] shows users with a higher average age (71.02 ± 13.94 years) when compared to this study (45.9 ± 23 years), which indicates a significant difference in the age profile of Brazilian users and a greater prevalence of physical disabilities among Brazilians aged 15 to 64 years of age, what is also shown in the last national census (especially for mild and moderate physical disabilities) [13].

The significant number of proxy respondents (39.65%) and specially caregivers (32.5%), might indicate that users also face barriers to communicate that have not been tackled by the provision of assistive products, as these are currently out of the scope of products and services offered by the Brazilian Public Healthcare System.

Abandonment, satisfaction, efectiveness and impacts of assistive products have also been assessed by Federici et al. [5] and others [14–17], who reiterate that there are no standard assessment tools to investigate the abandonment of assistive products. Several studies report that abandonment level are constantly around 30% [5,14,17,18]. A recent study perfomed telephone interviews with 749 Italian users of assistive products and demonstrated that 22.4% interrupted the use of their devices [5].

This study shows that 83.5% of users used at least one of the assistive products they have received, although only 70% of them actually used all of the products they have received. On the other hand, it also shows that 80.62% of the assistive products provided were been used at the time of the interview, an index that is similar to other studies [5,14,17,18].

Using these two different perspectives, one that focused on users and the other on the assistive devices provided to them, allows us to identify their complementarity. Although the usual focus, on the product, is informative, it is important to highlight that users need to receive special attention, once this is not a direct relation, specially when we consider the use of multiple devices.

When assessing the use of multiple devices, the major part (81.65%) of users receives up to three assistive products. Hence, this population requires a special attention by service managers, physcians and therapists who are involved in delivering services to assistive products users, mainly because of the alarming disparity in relation to their adherence to the use of assistive products.

A striking difference between the odds ratios on the use of assistive products of both groups (0.36 for those who received only one and 1.72 for those who received three) demonstrates that the profiles of these users can be remarkably different. Among those who received only one device, abandonment levels of shower wheelchairs (40%), folding frame wheelchairs with (35%) and without (33%) postural support devices (PSD) are meaningful. On the other hand, those who have received three assistive products have mostly abandoned upper limb prosthesis (67%) and crutches (50%). Abandonment levels for lower limb prostheses and walkers are high in both groups.

Moreover, the 1.99 average for assistive products provided by user and the 1.56 average use index (for those who received up to three devices) are significant and important when planing rehabilitation services and systems and the provision of assistive products in similar settings.

Reflecting the Brazilian context, where assistive products are often provided within rehabilitation treatment, this study demonstrates that there are remarkable differences between the use and abandonment levels of users who are still on rehabilitation treatment and those who have completed it. Completing rehabilitation treatment not only impacts abandonment, but also influences the

Table 5. Factors that impact on the abandonment of assistive products.

Products	Percentage of abandonment	Factors that influence abandonment	
Cane	16.67%	Positive	Personal: 50% Product: 33.3%
		Negative	Intervention: 16,7% Personal: 16.7%
Crutch	31.43%	Positive	Personal: 33.3% Intervention: 33.3%
		Negative	Personal: 33.3% Environmental: 11.1%
Folding frame wheelchair with PSD	15.53%	Positive	Personal: 30.8% Product: 26.9% Intervention: 26.8%
		Negative	Personal: 15.3% Environmental: 3.8% Others: 7.7%
Folding frame wheelchair without PSD	16.49%	Positive	Personal: 50% Personal: 33.4%
		Negative	Product: 8,3% Others: 8,3%
Lower limb orthosis (KAFO)	26.37%	Positive	Personal: 15% Intervention: 50% Personal: 35%
		Negative	Product: 5% Others: 5%
Lower limb orthosis (others)	18.39%	Positive	Personal: 24% Intervention: 48% Personal: 16%
		Negative	Product: 8% Environmental: 2% Others: 12%
Lower limb prosthesis	33.87%	Positive	Personal: 5% Intervention: 40% Personal: 30%
		Negative	Product: 15% Others: 10%
Rigid frame wheelchair with PSD	7.41%	Positive	Personal: 50%
		Negative	Product: 50%
Rigid frame wheelchair without PSD	10.00%	Positive	Personal: 100%
		Negative	0%
Shower wheelchair	13.19%	Positive	Personal: 48.1% Intervention: 18.5% Product: 14.8%
		Negative	Environmental: 11.1% Personal: 3.7%
Upper limb orthosis	23.26%	Positive	Personal: 28.6% Intervention: 31.8% Product: 19.1%
		Negative	Personal: 14.3% Environmental: 3.2% Others: 9,5%
Upper limb prosthesis	53.33%	Positive	Personal: 37.5% Intervention: 25%
		Negative	Personal: 25% Product: 12.5%
Walker	30.77%	Positive	Personal: 75% Intervention: 8.3%
		Negative	Environmental: 8.3% Others: 8.4%

chances to use assistive products once they diminish as time goes on.

This can be related to the fact that rehabilitation motivates the user and usually requires the user to actively use the device as part of his/her treatment, as well as for his/her training. Thus, short-term use is positively influenced by the rehabilitation treatment, as it increases the chance to use an assistive product (2.25).

Furthermore, long-term use is hindered by the completion of the rehabilitation treatment, once chances to use an assistive device diminish as time goes on (2.74 during the first year, 0.80 after the first year and 0.24 after the third year). This factor reiterates the need for follow up after discharge from the rehabilitation treament.

By means of a systematic follow up, service providers are able to identify changes in the physical and functional conditions of the user, on its actual and intended uses, on the maintenance of the product and even on the environments where the device has been used, besides identifying possible flaws in former steps of service provision, such as fitting or training on the safe and

effective use of the device [19]. All of these are important factors that diminish the chances to use assistive devices on the long-term and increase abandonment levels.

Another meanigful factor for users adherence to the use of assistive products, which would benefit from specific measures aimed at it by health and rehabilitation systems, services and professionals, is trying to promote (still during treatment) the perception of importance about the use of assistive products among their users, for chances to use assistive products are higher (3.84) among those who perceive it as significant.

The assistive products included in this study reflect the range of products offered by the Brazilian Public Healthcare System for users with physical disabilities and mobility impairments. Among them, upper and lower limb prostheses, walkers and crutches have an overall abandonment level of above the 30% index.

As per this study, the highest abandonment rates are associated with upper limb prosthesis (53.3%; mainly due to negative factors). This might be a result of upper limb prostheses which are still not functional. These users might function better using adaptors than using the products which are currently available. This gets remarkably clear once one notices that 37.5% of users indicate their own functional improvement as the reason for abandoning prostheses.

Secondly, lower limb prostheses have an abandonment level of 33.87%, mainly due to negative factors which are intervention-related and personal, such as inadequate size and difficulties in adapting to the use of prostheses. This might be related to problems in the assessment or prescription. Many times, stumps are short or the prescription was aimed at attending esthetic needs, which might not be a real demand by the user, especially after functional gains resulting from the rehabilitation treatment [20].

With reference to walkers and even to crutches, the evident nature of temporary use can be captured once one perceives that the factors contributing towards their abandonment levels are mostly positive and related to improvements in health and functioning conditions of users.

Positive factors are not only related to products whose use are temporary, such as walkers and crutches or even upper limb prostheses, as per the reasons explained before. Among the 13 assistive products included in this study, there are six for which at least 50% of factors are positive, especially wheelchairs.

Among wheelchairs, shower wheelchairs, rigid frame wheelchairs (both with and without PSD) and folding frame wheelchairs (without PSD) present low abandonment levels and make a significant contribution to the levels of positive abandonment factors, what might also reflect the temporariness of their use. However, these devices are not currently received by the Brazilian Public Healthcare System after they are no longer used, which is a missed opportunity to save resources.

Unfortunately, personal factors for the abandonment of assistive productives are not restricted to positive ones. Some personal factors cause a negative impact over the use of assistive products, such as difficulties in adapting to or accepting their use or even worst health and functioning status, which include new diseases and injuries. These factors are especially important for the use of prostheses, crutches, folding frame wheelchairs with postural support devices and Knee Ankle Foot Orthosis (KAFO).

In a recent publication [20], authors indicate that worsened health conditions are a major factor for the abandonment of assistive products, Nevertheless, curiously, this factor was not equally significant in this study.

Other possibly suggestive negative factors are those related to the intervention itself. There are several assistive products for which more than 25% of users reported an intervention-related factor as a determinant for its abandonment. Notably, intervention-related factors are appointed by more than 40% of users as having been determinant for abandoning lower limb prostheses (for the reasons explained before) and lower limb orthoses, especially KAFO.

Products themselves, if of poor quality, can be prematurely worn out or broken and thus be abandoned by their users. Factors such as deterioration, lack of adjustments and frequent damages are indicated by other authors [20–22] as possible causes for the lack of effectiveness and durability, which negatively affect their use [14]. Symptomatically, folding frame wheelchairs with PSD are the devices for which product-related factors are most important. Users frequently report that folding frame wheelchiars, as well as PSD, are rapidly worn out, hindering their use.

The users of folding frame wheelchairs with PSD are also among those that report to face environmental barriers to use their devices and hence to abandon it. Environmental factors are also significant for users of shower wheelchairs, walkers and crutches. It is worth mentioning that environmental barriers are found outdoors, on the streets, public buildings and transport system, and indoors, in users' homes (what is especially true for shower wheelchair users).

As previously stated, the health system and assistive products service provision themselves could be a barrier that prevent the use of assistive products [20,23]. In Brazil, the Public Healthcare System is the main provider of assistive products, what simplifies access to assistive products and services. Out of the Public Healthcare System, there are other possible alternatives, for which users often appeal when they go to court to ensure their access to assistive products. However, this study only includes users who were appropriately serviced by the Brazilian Public Healthcare System, that guarantees some level of homogeneity among them.

This study also has some twolimitations. First, the number of survey respondents is only a fraction of the total number of service users, hence limiting the number of users in some categoriesand second, the assistive products covered by the survey are limited to mobility devices, which is due to the kind of rehabilitation services provided and the target-audience of the Institute.

The geographical area covered by the Institute, although restricted to the neighboring cities of Sao Paulo, ensures a service coverage that embraces an expressive population within one of the largest cities and metropolitan areas in the world (nearly 22 million people in 2016).

This study's approach to the theme is certainly innovative in our region. The sample size and the face-to-face interviews all contribute to sound data that aids in strategic decision making on the provision of assistive products.

Conclusion

83.5% of the users use at least one of the assistive products they have received. Moreover, the general abandonment level of assistive products was 19.38%. The majority of users (81.65%) received up to three assistive products, resulting in a ratio of 1.99 products and provided and of 1.56 products effectivelly used per user.

The simultaneous use of mutiple assistive products, users perception on the importance of using them and completing the rehabilitation treatment were found to impact on the short- and long-term use of assistive products.

Rigid and folding frame wheelchairs, with and without PSD, as well as shower wheelchairs, presented the lowest abandonment levels, followed by canes and lower limb orthoses. Upper limb

orthoses, KAFO, walkers, crutches and lower and upper limb prostheses all presented higher abandonment levels. We believe these data could assist informing decision making on assistive products provision and servicing in similar seetings.

From a systems perspective, this study provides information on the expected demand for assistive products for people with physical disabilities in a developing country. It offers a glance about user's characteristics, the range of products that could be made available to persons with mobility impairments and inputs to estimate resource use and financial impact.

It also highlights the impact of associating rehabilitation treatment and assistive products provision and of following up on users for short- and long-term use. The study also demonstrates the importance of considering the need for reusing assistive products for the environmental and financial sustainability of assistive technology service provision.

At service delivery level, it shed light on the assistive products that require special attention by service personnel for their safe and effective use, meaning greater focus on user assessment, prescription, training and follow up. It also offers an insight into the importance of considering the simultaneous use of multiple devices when trying to meet the often complex needs of users. Reiterating a user-centered approach, the study also stresses the need to consider their awareness and individual standing with reference to the assistive products provided.

Questions used in the interviews

- Are you a patient, former patient or caregiver/personal assistant?
- What is the user's gender? (male/female)
- What is the user's date of birth?
- Is the user currently under rehabilitation treatment? (yes/no)
- When did the user startrehabilitation treatment?
- When has the user stoped rehabilitation treatment?
- Has the user received any kind of device to assist his/her rehabilitation? (yes/no)
- What devices has the user received? (upper limb prosthesis, lower limb prosthesis, upper limb orthosis, lower limb orthosis, KAFO, rigid frame wheelchair without PSD, rigid frame wheelchair with PSD, folding frame wheelchair without PSD, folding frame wheelchair with PSD, shower wheelchair, walker, crutches and cane)
- Does the user use the devices he/she has received? (yes/no)
- What devices does the user use? (upper limb prosthesis, lower limb prosthesis, upper limb orthosis, lower limb orthosis, KAFO, rigid frame wheelchair without PSD, rigid frame wheelchair with PSD, folding frame wheelchair without PSD, folding frame wheelchair with PSD, shower wheelchair, walker, crutches and cane)
- Does the user have any difficulties in using the devices received? (yes/no)
- What difficulties does the user have in using the devices received? (open answer)
- If the user doesn't use some of the devices he/she received, why is it? (it does not perform the intended function, it does not fit the user, user conditions improved and he/she no longer needs the device, user conditions worsened and the device does not meet his/her current needs, the user has not adapted to use the device, the user cannot wear the device, the user does not accept the device, the user feels discriminated when using the device, the device was stolen, the device was lost, the device is broken or worn out, there are physical barriers that prevent the use of the device and open answer)
- In a scale that goes from 1 to 5, in which 1 is not important and 5 is very important, how important is the use of assistive devices during rehabilitation treatment? (1. not important, 2. a bit important, 3. indifferent, 4. Important and 5. very important)

Disclosure statement

No potential conflict of interest was reported by the authors.

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